```
import matplotlib.pyplot as plt

training_data = [
["High", "Loamy", "Medium", "Warm", "Organic", "Yes"],
["High", "Loamy", "Medium", "Warm", "Organic", "Yes"],
["High", "Clay", "High", "Warm", "Organic", "No"],
["Medium", "Loamy", "Medium", "Warm", "Organic", "Yes"],
["Low", "Loamy", "High", "Cool", "Chemical", "No"],
["High", "Loamy", "Medium", "Warm", "Chemical", "Yes"],
["Medium", "Clay", "Low", "Cool", "Organic", "No"],
["High", "Loamy", "Medium", "Warm", "Organic", "Yes"]
]

attributes = [
"Sunlight",
"Soil",
"Water",
"Temperature",
"Fertilizer"
]

def find_s(data):
hypothesis = None

for row in data:
target = row[-1]
example = row[:-1]

if target == "Yes":

if hypothesis is None:
hypothesis = example.copy()
print("Initial hypothesis:", hypothesis)

else:
for i in range(len(hypothesis)):
if hypothesis[i] != example[i]:
hypothesis[i] = "?"

print("Updated hypothesis:", hypothesis)

return hypothesis
```

```
final_hypothesis = find_s(training_data)

print("\nFINAL MOST SPECIFIC HYPOTHESIS")
print("=======================================")

for attribute, value in zip(attributes, final_hypothesis):
print(f"{attribute:15} : {value}")

specificity = [
1 if value != "?" else 0
for value in final_hypothesis
]

plt.figure(figsize=(9, 5))

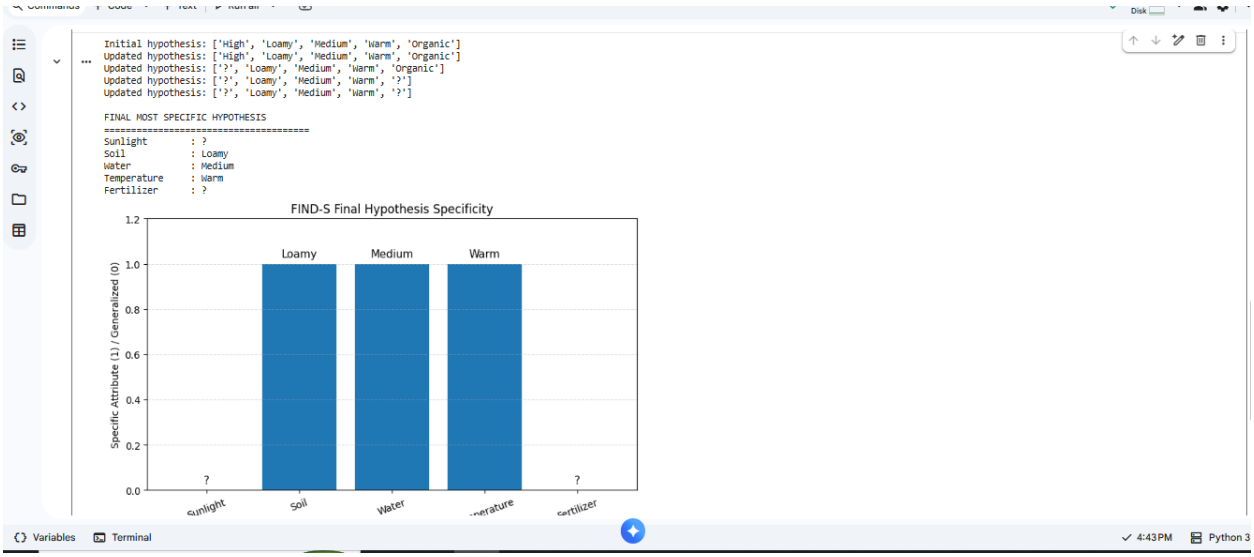
bars = plt.bar(attributes, specificity)

plt.title("FIND-S Final Hypothesis Specificity")
plt.xlabel("Attributes")
plt.ylabel("Specific Attribute (1) / Generalized (0)")
plt.ylim(0, 1.2)

for bar, value in zip(bars, final_hypothesis):
plt.text(
bar.get_x() + bar.get_width() / 2,
bar.get_height() + 0.03,
value,
ha="center",
fontsize=11
)

plt.xticks(rotation=20)
plt.grid(axis="y", linestyle="--", alpha=0.4)
plt.show()
```

output


Initial hypothesis: ['High', 'Loamy', 'Medium', 'Warm', 'Organic']
Updated hypothesis: ['High', 'Loamy', 'Medium', 'Warm', 'Organic']
Updated hypothesis: ['?', 'Loamy', 'Medium', 'Warm', 'Organic']
Updated hypothesis: ['?', 'Loamy', 'Medium', 'Warm', '?']
Updated hypothesis: ['?', 'Loamy', 'Medium', 'Warm', '?']
FINAL MOST SPECIFIC HYPOTHESIS
Sunlight : ?
Soil : Loamy
Water : Medium
Temperature : Warm
Fertilizer : ?
FIND-S Final Hypothesis Specificity
Specific Attribute (1) / Generalized (0)
Loamy
Medium
Warm
Sunlight
Soil
Water
Variables
Terminal
4:43PM
Python 3